AQA 3.1.4 and 3.1.10 Equilibrium constants K_c and K_p

AQA A-level Chemistry · Revision questions · Sheet 7 of 9

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

Sheet 7 of 9

20 marks · about 25 minutes

Key information

- Find equilibrium amounts with a start / change / equilibrium table using the mole ratio in the equation
- K_c: divide each amount by the volume in dm^3 before substituting (unless the volumes cancel)
- Only temperature changes the value of K_p or K_c; pressure, concentration and catalysts do not
- Exothermic forward reaction: raising temperature lowers K; endothermic: raising temperature increases K
- Units: substitute the units into the expression and cancel (e.g. kPa^{-2}, $mol^{-1}\ dm^3$ or no units)

Section A: Short-answer questions

01 A gas-phase reaction is at equilibrium. When the pressure is increased the yield of product decreases. State what can be deduced about the chemical equation for this equilibrium.

[1 mark]

02 Deduce the units of K_p for $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ with partial pressures in kPa.

[1 mark]

03 An equilibrium mixture contains three gases A, B and C. The mole fraction of A is 0.35 and the mole fraction of B is 0.45. Calculate the mole fraction of C.

[1 mark]

04 An equilibrium mixture contains gases with partial pressures of 25 kPa, 60 kPa and 15 kPa. State the total pressure.

[1 mark]

Section B: Longer questions

05 Methanol can be manufactured in a reversible reaction as shown by the equation.

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

05.1 State and explain the effect of using a catalyst on the yield of methanol in this equilibrium.

[2 marks]

05.2 Give an expression for the equilibrium constant (K_c) for this reaction.

[1 mark]

05.3 A mixture of carbon monoxide and hydrogen was allowed to reach equilibrium in a container of volume 250 cm^3 at temperature T. At equilibrium, the mixture contained 0.340 mol of carbon monoxide, 0.190 mol of hydrogen and 0.0610 mol of methanol. Calculate the value of K_c (in $mol^{-2}\ dm^6$) for this reaction at temperature T.

[3 marks]

05.4 Methanol decomposes on heating: $CH_3OH(g) \rightleftharpoons CO(g) + 2H_2(g)$. Use your answer from the previous part to determine the value of K_c for this equilibrium at temperature T. State the units for this value of K_c. (If you were unable to complete the calculation, assume $K_c = 0.825\ mol^{-2}\ dm^6$. This is not the correct value.)

[2 marks]

06 The equation shows the final stage in the production of methanol.

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

06.1 20.1 mol of carbon monoxide and 24.2 mol of hydrogen were placed in a sealed container. An equilibrium was established at 600 K. The equilibrium mixture contained 2.16 mol of methanol. Calculate the amount, in moles, of carbon monoxide and of hydrogen in the equilibrium mixture.

[2 marks]

06.2 A different mixture of carbon monoxide and hydrogen was allowed to reach equilibrium at 600 K. At equilibrium, the mixture contained 2.76 mol of carbon monoxide, 4.51 mol of hydrogen and 0.360 mol of methanol. The total pressure was 630 kPa. Calculate a value for the equilibrium constant, K_p, for this reaction at 600 K and state its units.

[6 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	More moles of (gaseous) products than (gaseous) reactants	*Source: 7405/1 Jun 2022 Q01.2* Allow molecules; do not accept atoms	1
02	kPa^{-2}	*Source: Practice question (written for this set)*	1
03	0.20	*Source: Practice question (written for this set)*	1
04	100 kPa	*Source: Practice question (written for this set)*	1
05.1	M1 no effect (on yield) M2 increases rate of both forward and reverse reactions equally / by the same amount	*Source: 7404/2 Jun 2018 Q08* CE = 0 if yield changes	2
05.2	$K_c = [CH_3OH] \div [CO][H_2]^2$	Must be square brackets	1
05.3	M1 divides moles by volume (0.250) M2 $K_c = 0.244 \div (1.36 \times 0.762^2)$ M3 0.311	If volume not used, allow M3 only for 4.97	3
05.4	M1 $1 \div 0.311 = 3.22$ M2 $mol^2\ dm^{-6}$	M1 = 1.21 if the alternative value is used	2
06.1	CO = 17.9 mol H_2 = 19.9 mol	*Source: 7405/1 Jun 2017 Q09.2–09.3* Allow 17.94 and 19.88	2
06.2	M1 $K_p = p(CH_3OH) \div (p(CO) \times p(H_2)^2)$ M2 total moles of gas = 2.76 + 4.51 + 0.36 = 7.63 M3/M4 p(CO) = 2.76/7.63 × 630 = 228 kPa; $p(H_2)$ = 4.51/7.63 × 630 = 372 kPa; $p(CH_3OH)$ = 0.36/7.63 × 630 = 29.7 kPa M5 $K_p = 29.7 \div (228 \times 372^2) = 9.4(1) \times 10^{-7}$ M6 kPa^{-2}	All 3 partial pressures = 2 marks; 2 correct = 1 mark Allow 9.39 to 9.50×10^{-7}; 9.41×10^{-13} if Pa used (units Pa^{-2}) If no total moles calculated, can only score M1 and M6	6